

Overview

This report analyzes how six energy and utility stocks (CEG, FANG, FRO, NEE, AROC, and WDS) move in relation to four major commodities (WTI Crude, Natural Gas, Gold, and Copper) using three years of weekly return data across 156 observations. The clearest finding is that WTI Crude is the only commodity that consistently matters across the portfolio. FANG is the most oil-driven name, with roughly half of its weekly return variance explained by crude prices alone, making it effectively an oil proxy. WDS follows closely, with additional copper exposure consistent with its LNG and oil production mix. AROC, despite being a midstream infrastructure business, still shows meaningful oil linkage because upstream drilling activity cycles drive compression demand. FRO has a real oil beta as well, though tanker-specific dynamics like day rates and fleet supply account for the majority of its moves. The two utilities, CEG and NEE, sit in a completely different category. NEE has near-zero commodity sensitivity with 96% of its variance driven by company-specific factors and interest rate dynamics. CEG is similar. These two names are genuine diversifiers within the portfolio and do not respond meaningfully to commodity price moves. The biggest surprise in the data is Natural Gas. Despite AROC being a gas infrastructure company, Natural Gas spot prices have zero statistically significant relationship with any of the six stocks. The portfolio has no effective Natural Gas price exposure. AROC earns on throughput volumes under long-term contracts, not the commodity price itself. If there is a directional view on gas, the current holdings will not express it.

Data & Methodology

Weekly data was chosen over daily to smooth out day-to-day noise that does not reflect genuine economic relationships. Returns are calculated as the percentage change in price from one week to the next, expressed in log form to ensure consistency in comparisons across different assets. The analysis ran in three stages: correlation, covariance, and a statistical test. The first stage looked at correlation and covariance across all ten assets. Statistical tests are applied to determine which correlations are strong enough to be considered real rather than random. The second stage ran a regression for each stock against all four commodities simultaneously. The regression is a way of isolating how much each commodity independently contributes to a stock's returns, holding the others constant. This produces a beta for each commodity, which can be thought of as a sensitivity score. A beta of 0.75 for WTI Crude means that, on average, a 1% move in oil is associated with a 0.75% move in that stock. Before interpreting these betas, a check was performed to confirm that the four commodities are not too correlated with each other, which would make the individual betas unreliable. They are not, so each beta can be read on its own terms. Finally, the third stage looked at whether these relationships have been stable over time using one-year rolling windows and broke down how much of each stock's total price movement is explained by commodities versus factors specific to that company. A clustering analysis was also included, which grouped the stocks and commodities together based on how similarly they behave, providing a visual summary of which names are genuinely related and which are not.



Data Preparation

- Collecting Financial Data
- Log-Return Calculations



Measuring Relationships

- Finding correlation and covariance
- Regression Analysis
- Multicollinearity Check



Stress Testing the Results

- Rolling Correlations
- Variance Decomposition
- Clustering

Results

The correlation and covariance matrices confirm that WTI Crude is the dominant commodity factor across the portfolio. FANG shows the strongest oil correlation at 0.71, followed by WDS at 0.58, AROC at 0.42, and FRO at 0.38. All four are statistically significant at the 1% level. CEG shows a modest but significant correlation with WTI at 0.21, while NEE is essentially flat at 0.01. Natural Gas produces no significant correlation with any stock in the set. Gold shows mild but statistically significant correlations with NEE (0.20) and WDS (0.16), and Copper registers broad but moderate significance across most names. The OLS regressions sharpen these relationships into actionable betas. FANG carries a WTI beta of 0.75, meaning a 1% move in crude is associated with a 0.75% move in FANG on average. WDS comes in at 0.54 and AROC at 0.46, both significant at the 1% level. FRO has a WTI beta of 0.50, also highly significant. CEG is the only name where WTI is significant at the 5% level rather than 1%, with a beta of 0.27. NEE shows no significant beta against any commodity, with an F-statistic p-value of 0.42 indicating the model has no explanatory power for that stock. The R-squared value answers one simple question: out of all the weekly price movement in a stock, how much of it can be explained by commodity prices? FANG sits highest at 52%, meaning just over half of its weekly moves can be traced back to oil, gas, gold, and copper. WDS follows at 40% and AROC at 20%. FRO comes in at 18%, CEG at 8%, and NEE at just 4%. The remaining percentage in each case is driven by factors specific to the company itself, things like earnings, management decisions, or industry dynamics that have nothing to do with commodity markets. The variance decomposition confirms that WTI Crude accounts for nearly all the commodity-explained portion in every case, with Natural Gas contributing essentially nothing and Gold and Copper making only minor contributions. The rolling 52-week correlations show that these relationships have been broadly stable over the three-year period, though FANG's WTI correlation tightened further during periods of elevated oil volatility and loosened slightly heading into 2026. NEE and CEG remain flat and near zero throughout, with no periods where commodity correlations spiked meaningfully. The hierarchical clustering output separates the portfolio cleanly into two groups: an oil-beta cluster containing FANG, WDS, WTI Crude, AROC, and FRO, and a separate cluster containing NEE, CEG, Gold, and Copper, reflecting the fundamental business model differences between the two groups.

Key Findings

WTI Crude Is the Dominant Factor: Oil is the only commodity that consistently matters across the portfolio, significant for five of six stocks. FANG is the most sensitive with a beta of 0.75, followed by FRO (0.50), WDS (0.54), and AROC (0.46).

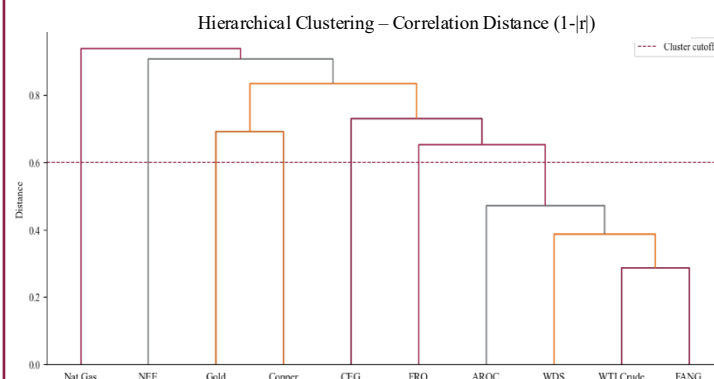
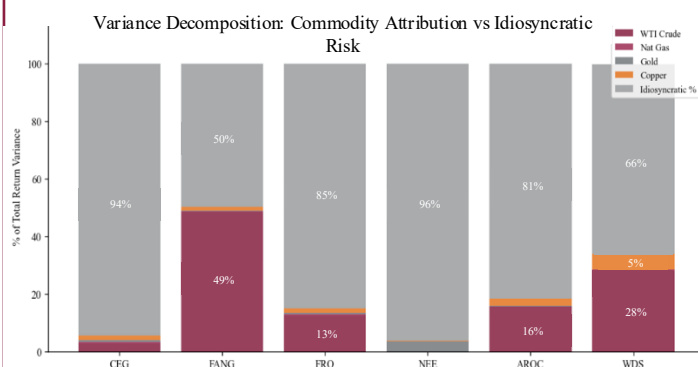
Natural Gas Is Disconnected From Everything: Despite holding two gas-linked names, the portfolio has zero statistically significant Natural Gas exposure. AROC earns on output volumes under fixed contracts, not the spot gas price, so its stock does not respond to gas price moves.

Gold Reflects Interest Rates, Not Commodity Exposure: Gold shows a mild but significant correlation with NEE and WDS, not because of any operational link, but because all three assets respond to the same underlying force, falling real interest rates. This is a shared macro sensitivity, not a commodity relationship.

Copper Is a Global Growth Signal: Copper shows broad but moderate significance across most names, most clearly in WDS. It is best read as a global economic activity indicator, when the world economy runs hot, energy demand follows, rather than a direct operational exposure.

Two Distinct Clusters: The portfolio splits cleanly into an oil-beta group (FANG, WDS, AROC, FRO) and a rate-sensitive utility group (CEG, NEE). The two groups move largely independently of each other, meaning CEG and NEE are providing genuine diversification against the commodity risk in the other four names.

Idiosyncratic Risk Dominates: Even in FANG, the most commodity-driven stock, half of its weekly price movement is still driven by company-specific factors. For NEE that figure is 96%. Commodity hedges would only partially protect this portfolio, most risk in most names sits at the company level, not the commodity level.



Recommendation

Given the data, the portfolio is well-structured but carries concentrated oil risk through FANG and WDS, which together account for the majority of commodity-driven variance. If the energy sector has a forward-looking view on crude, the current positioning makes sense, and FANG is the cleanest expression of that trade. If the outlook on oil softens, trimming FANG first would be the most efficient way to reduce commodity exposure given its beta of 0.75. CEG and NEE serve as natural portfolio stabilizer and are worth holding for their rate-sensitivity and diversification value, particularly in an environment where interest rates are expected to decline. The one gap worth addressing is Natural Gas, if there is a view on gas prices, none of the current holdings will express it, and rotating into a pure-play gas E&P name would be the most direct way to fill that exposure.

Implications

The core implication is that this portfolio is essentially a leveraged bet on the oil cycle, with a built-in hedge through CEG and NEE. When crude is rallying, FANG, WDS, AROC, and FRO will likely outperform. When crude sells off, those same names will absorb the most damage. Understanding that dynamic is important for sizing and risk management. The Natural Gas gap is the most actionable structural issue. The portfolio touches the gas industry through AROC but captures none of the price upside or downside. Any macro view on gas prices is effectively going unexpressed. The high idiosyncratic risk across most names also implies that stock selection and fundamental research matter more here than commodity timing. Having a correct view on oil is helpful, but it is only part of the equation. Each company still has enough company-specific variance that picking the right names within the energy space remains the primary driver of returns.

Conclusion

This analysis set out to understand how a portfolio of six energy and utility stocks relates to four major commodity markets, and the results paint a clear and actionable picture. WTI Crude is the dominant external force shaping returns across the portfolio, with meaningful and statistically significant relationships running through FANG, WDS, AROC, FRO, and to a lesser extent CEG. The portfolio is, at its core, an oil-cycle portfolio, and performance will largely reflect how crude moves over any given period.

At the same time, the analysis reveals important nuance beneath that headline. Natural Gas, despite its relevance to several of the underlying businesses, has no statistically detectable influence on any stock in the set. The portfolio is exposed to the gas industry operationally but not financially, a distinction that matters if gas prices become a key macro theme. Gold and Copper both appear as secondary signals, though neither represents a direct commodity exposure. Gold can be better understood as a proxy for the interest rate

environment, while Copper reflects sensitivity to the broader global growth cycle.

The utility names, CEG and NEE, stand apart from the rest of the portfolio in a meaningful way. Their near-zero commodity sensitivity and high idiosyncratic character make them genuine diversifiers, providing stability when energy markets are volatile and responding instead to a different set of macro drivers centered on interest rates and regulatory dynamics.

Perhaps the most important takeaway for portfolio management is that even in the most commodity-driven names, a significant portion of return variance is driven by company-specific factors that no commodity hedge can address. This reinforces the value of rigorous fundamental research alongside macro and commodity analysis. The two are not substitutes for each other. A portfolio like this one requires both a view on the oil cycle and a view on each individual business to be managed effectively.

Rolling 52-Week Correlations: Each Stock vs Commodities

